HDOC Day-50

Aim: Write a C program using pointers and a separate function majority_element() to find an element occurring more than size/2 times. Print -1 if none exists.

C Program:

```
#include <stdio.h>

void majority_element(int *nums, int size)
{
    int i, j, count;

    for (i = 0; i < size; i++)
    {
        count = 0;
        for (j = 0; j < size; j++)
        {
            if (*(nums + i) == *(nums + j))
                count++;
        }
        if (count > size / 2)
        {
            printf("Majority element: %d\n", *(nums + i));
            return;
        }
    }
    printf("-1\n");
}

int main()
{
    int size, i;
    printf("Enter size of array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size\n");
        return 0;
    }

    int nums[size];
    printf("Enter array elements: ");
    for (i = 0; i < size; i++)
        scanf("%d", &nums[i]);

    majority_element(nums, size);
    return 0;
}
```

Sample Outputs:

Example 1: Size and elements: 5 | 3 3 2 3 3
Output: Majority element: 3

Example 2: Size and elements: 7 | 2 2 1 1 1 2 2
Output: Majority element: 2

Example 3: Size and elements: 6 | 1 2 3 2 3 3
Output: -1